



Evaluation of Cue Lure Longevity and Effectiveness against *Bactrocera cucurbitae* (Coquillett) in Bitter Gourd Fields

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Article Received 01-04-2026, Article Revised 28-6-2026, Article Accepted 20-08-2026

ABSTRACT:

The different hosts available throughout the world is a major source of occurrence of insect pest population. The distributions of fungivorous pest have variety of host plant. The forecasting and prediction are the core of the integrated management of these harmful fruit eating pests. The melon fruit fly *Bactrocera cucurbitae* (Coq.) is considered as a serious pest of cucurbit vegetables throughout the world. The management of the pest needs a proper monitoring before application of any management tool. Cue-lure of *Bactrocera cucurbitae* (Coq.) in bitter gourd field" was used for monitoring *B. cucurbitae* population in different interval days i.e. after 15days, 25days, 35days and 45days were conducted under field conditions during 2017. *B. cucurbitae* as a serious pest of vegetables and fruits in Pakistan. Male flies were attracted to cue lure, which was placed in traps, on different interval days. The peak population was observed during first week of July. The highest population (109.04 ± 14.57) was observed after 15 days application of cue lure followed by 108.26 ± 14.47 after 25 days, 99.23 ± 13.26 after 35 days and 86.98 ± 11.62 after 45 days application of cue lure respectively. Correlation with abiotic factors i.e., temperature and relative humidity (%) had highly significant ($P = <0.001$) positive correlation on the population incidence of melon flies.

Keywords: Bitter gourd, Cuelure, *Bactrocera cucurbitae*,

INTRODUCTION

Bitter gourd (*Momordica charantia* L.) is an important Kharif season vegetable crop belongs to family Cucurbitaceae. The major cucurbitaceous vegetable are grown throughout the Asia, especially in Pakistan, China, India, Sri Lanka and Bangladesh (Behera, 2004; Tahir and Haider, 2005; Akram et al., 2010). Mostly vegetables are used as nutritional purpose thus people use vegetables as daily energy diet. The vegetables are also used as remedy for human diseases. The bitter gourd has great medicinal value, Charantin, a chemical present in bitter gourd is used to reduce blood sugar for diabetic patients and is also rich in carbohydrates (Rani et al., 2014). The vegetables play a leading role in our daily life but unfortunately vegetables are short duration crop and easily under attack of harmful insect pest population. The harmful flies are very serious problem for fresh fruits and vegetables in the world. The fruit flies are not only causing direct damage to vegetables but they also risk for vegetable trade (Joomaye, 2000). The fruit flies are destructive insect pests of cucurbit and other fruits, nevertheless, among them *B. cucurbitae* is

the most injurious pests of bitter gourd, pumpkins, bottle gourd, Indian squash and other fleshy fruits (Bharathi et al., 2004). The *B. cucurbitae* is a polyphagous fruit fly pest that damages more than 125 species of plants, frequently in the Cucurbitaceae, like bitter guard, cucumber, pointed gourd, bottle gourd and squash. The larvae of melon fruit fly mostly damage inside the fruit and also fruit pulp. Mostly infested fruits and flowers fall down and remaining do not developed properly. The damage often cause dramatic reduction in yield (Dhillon et al., 2005). The punctures done by females without eggs and brown resinous secretion makes inedible and also reduce the quality the market value of vegetables crop (Gogi et al., 2009). The indiscriminate use of chemical insecticides has increase to resistance of noxious fruit flies, heavy resurgence to infestation by some insect species because of the demolition of natural enemies i.e., predators and parasitoids (Sarwar et al., 2013). The males of many fruit fly species are highly attracted to plant chemicals or their close chemical equivalents. These chemicals are called as parapheromones or male lure. The chemical lure are

used to control and eradicated fruit flies. There are two famous lures used as male lures such as methyl-eugenol (ME) (4-[allyl-1,2-dimethoxybenzene]) and cue-lure (4-[4-50 acetoxypheyl]-2-butanone), but other lures occur methyl-eugenol 51 is found commonly in nature, while cue-lure is a synthetic chemical 52 with natural analogues (Kumaran et al., 2014). The cue lure and methyl eugenol baited traps are frequently used against fruit flies. The methyl eugenol highly male attractive lures for *B. dorsalis* and cue-lure is the best attractant for male flies of melon fruit flies (Khan et al. 2003). The excess use of long residual pesticides has displayed side effects on soil and environment. Only insecticide remedy is not enough for controlling the notours fruit flies. It is very necessary to manage the fruit fly population through environmental free control measures. The sex attractant (lure) is very effective tool for killing of male flies (Mahmood et al., 2002). The aim of the present study is to evaluate the effectiveness of cue lure against *B. cucurbitae* in bitter gourd vegetable crop in order to control pest population.

MATERIALS AND METHODS

The experiment "Assessment of cue lure persistency for *Bactrocera cucurbitae* (Coq.) in bitter gourd field was conducted in the field at Rajput Vegetable Farm Bhria City District Naushahro Feroze. The experiment was conducted during kharif season 2017. The four acres area was recommended for this study. The plant spacing (hill x row) of 0.5x2 m was maintained. The 16 traps were baited with cue lure in bitter gourd field. The Randomized Complete Block Design (RCBD) were used with four treatments i.e., T1 =15days, T2 =25days, T3= 35days, T4 = 45days. The collected data were repeated four times. Each cue-lure baited traps were installed at 4 ft height above ground level. Cu-lure was proved from Nuclear Institute of Agriculture (NIA) Tandojam. The traps were baited with Cue-lure and insecticide (95% cue lure and 5% insecticide) deployed selected locations from 4th week of April 2017 to 1st week of August 2017. Male flies were attracted to lure and quickly killed by the insecticide on the cotton wick. The captured male flies were collected and counted on weekly basis. The effect of a biotic factors i.e., temperature and relative humidity on the population of pest were also recorded. The temperature and relative humidity were also recorded.

Statistical analysis: The data were analyzed using software Statistix 8.1 (Statistix, 2006).

RESULTS AND DISCUSSION

The Fruit flies are among the world's noxious insect pests of fruits, vegetables and other valuable important agricultural crops. There are many species, of fruit flies harmful for more valuable fresh fruits out of which majority of species belong to Dacine fruit flies (Demeyer et al., 2010). The *B. cucurbitae* is considered as a serious insect pest of fruit and

vegetables in Pakistan. *B. cucurbitae* is considered as a noxious insect nuisance of fruit and vegetables. It has wide range of cucurbit vegetable host plants for their survival. The present studies were carried out during the year of 2017 "Assessment of cue lure persistency for *Bactrocera cucurbitae* in bitter gourd field". The experiment was conducted at the field of Rajput Vegetable Farm Bhria City District Naushahro Feroze. The analysis of data showed that effects of cue lure application intervals were significant ($P<0.05$) and the results regarding weeks of the observation were also observed significant ($P<0.05$) on Catching of melon fly (*Bactrocera cucurbitae*), however the result about that the interaction was non-significant ($P>0.05$). The results in Table 1 were indicates that the melon fly (*B.cucurbitae*) appeared during the 4th week of April up to 1st week of August when the crop was harvested at last stage. During initial period of the Crop (bitter gourd) vegetable crop (After 15 days) interval , the activity of the melon fly remained very low with (12.00 ± 1.29) during 1st observation 4th week of April after that flies activity reached to its peaks i.e., (199.50 ± 10.82) , males/trap/week in 1st week of July . Then, the appearance was slowly declined with (95.25 ± 6.13) during the last week of observation. Than after population was also noted on bitter gourd (After 25 days) interval of cue lure. However, during 1st observation of the population incidence recorded was (9.50 ± 1.04), afterwards the population activity of the pest was increased to its high peak i.e., (198.50 ± 10.02) males/trap/week during 1st week of July. Then after population trend of flies very decreased with (94.50 ± 5.98) males/trap/week during final observation. Thereafter that the population activity of pest was recorded/ noted on bitter gourd (After 35 days) interval from 4th week of April with population of (7.25 ± 0.48) males/trap/week. The population increase to (187.25 ± 11.08) during 1st week of July. Thereafter, its population declined and reached to (84.00 ± 6.10) up to harvest of crop in first week of August. While appearance of *B.cucurbitae* was recorded on bitter gourd after (45 days) interval from 4th week of April with population of (6.75 ± 0.48) males/trap/week. The population reached to its maximum (171.50 ± 10.84) during 1st week of July. Thereafter, its activity decreased to (54.25 ± 5.25) up to harvest of crop in first week of August. It is concluded from the data that adult melon fruit flies were captured through cue lure traps (Sex pheromone traps) during various different interval days. Present findings are agreed with Kumar et al. (2006) argued and described that the melon fly, *B. cucurbitae* is one of the major frugivorous pest and highly damage to cucurbitaceous vegetables and fruits in many regions of the world. Further Figure 1 showed that highest population 109.04 ± 14.57 was recorded after 15 days (cue lure application), followed by 108.26 ± 14.47 (after 25days cue lure application), 99.23 ± 13.62 (after 35 days cue lure application) and 86.98 ± 11.26 after

45 days cue lure application respectively. Overall mean population of *B. cucurbitae* observed on various dates are given in Figure 2 showed that initially population remained very low (8.88 ± 1.20) at the start of season. However, a continuous increase in

population was observed through the growth of bitter gourd vegetables reaching at its peak on 1st week of July 2017 that was (189.19 ± 6.52). Afterwards, population started declining towards the harvesting of bitter gourd vegetable crop.

Table 1. Weekly population of *Bactrocera cucurbitae* (Coq.) under the influence of cue-lure application in bitter gourd during kharif season 2017

Months	Weeks	15 days	25 days	35 days	45 days
April	4 th week	12.00 ± 1.29^{YZa}	9.50 ± 1.04^{Za}	7.25 ± 0.48^a	6.75 ± 0.48^a
	1 st week	$22.25 \pm 2.06^{W-Za}$	$22.00 \pm 1.87^{W-Za}$	$17.25 \pm 1.89^{X-Za}$	$15.50 \pm 1.94^{Y-Za}$
May	2 nd week	$33.25 \pm 2.46^{U-X}$	$32.25 \pm 2.78^{V-Y}$	$28.00 \pm 3.94^{V-Z}$	$21.50 \pm 1.94^{W-Za}$
	3 rd Week	$54.25 \pm 3.09^{Q-T}$	$53.50 \pm 3.48^{R-U}$	$47.50 \pm 4.92^{S-V}$	$40.00 \pm 4.10^{T-W}$
	4 th week	$74.50 \pm 5.17^{O-Q}$	$72.50 \pm 5.63^{P-R}$	$62.50 \pm 5.14^{Q-S}$	$56.25 \pm 5.33^{Q-T}$
June	1 st week	103.00 ± 7.36^{MN}	102.00 ± 7.36^{MN}	95.25 ± 6.06^{MN}	$85.50 \pm 5.63^{N-P}$
	2 nd week	124.75 ± 8.17^{KL}	$125.50 \pm 8.99^{J-L}$	114.50 ± 9.54^{LM}	102.25 ± 6.55^{MN}
	3 rd week	$152.50 \pm 9.21^{E-H}$	$152.00 \pm 9.25^{E-I}$	$141.75 \pm 10.63^{G-K}$	$128.75 \pm 9.46^{J-L}$
	4 th week	$178.75 \pm 9.69^{B-D}$	177.00 ± 9.21^{CD}	$161.50 \pm 10.66^{D-G}$	$145.50 \pm 9.65^{F-J}$
	1 st week	199.50 ± 10.82^A	198.50 ± 10.02^{AB}	$187.25 \pm 11.08^{A-C}$	$171.50 \pm 10.84^{C-E}$
July	2 nd week	$181.75 \pm 9.19^{A-D}$	$181.25 \pm 9.01^{A-D}$	$164.00 \pm 10.64^{D-F}$	$154.50 \pm 9.26^{E-G}$
	3 rd week	$163.50 \pm 9.63^{D-F}$	$163.25 \pm 8.98^{D-F}$	$153.25 \pm 10.16^{E-H}$	$133.75 \pm 7.39^{H-L}$
	4 th week	$131.25 \pm 8.39^{J-L}$	$132.00 \pm 7.39^{J-L}$	$125.25 \pm 8.87^{J-L}$	101.75 ± 6.34^{MN}
August	1 st week	95.25 ± 6.13^{MN}	$94.50 \pm 5.98^{M-O}$	84.00 ± 6.10^{NP}	$54.25 \pm 5.25^{Q-T}$

Note: Means were donated by same alphabets are not significant different from each other

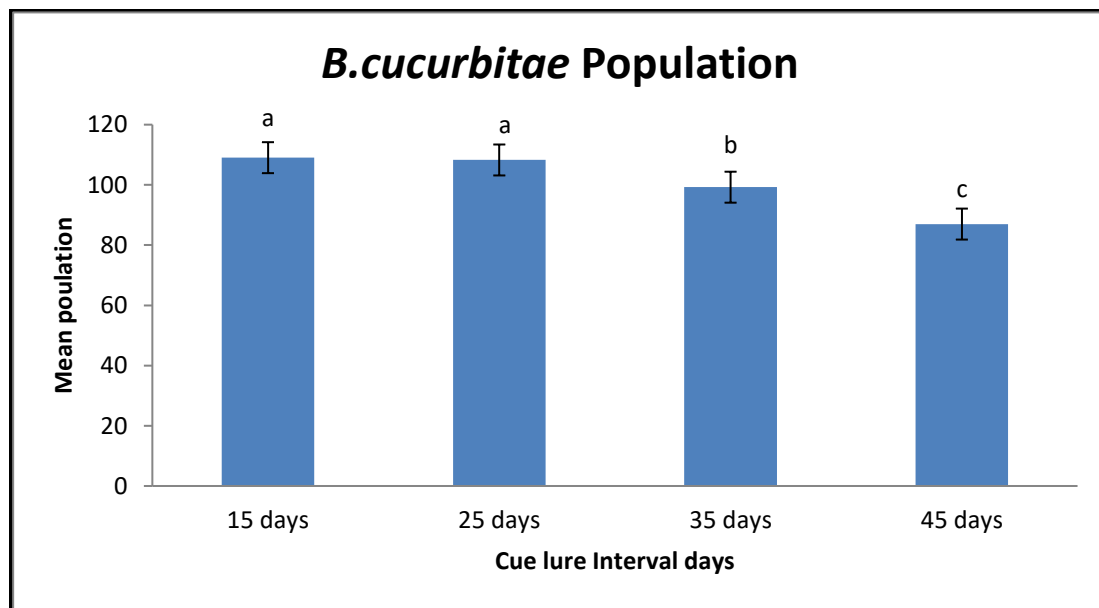


Figure 1. Population of *B. cucurbitae* on bitter gourd after cue-lure application during season 2017

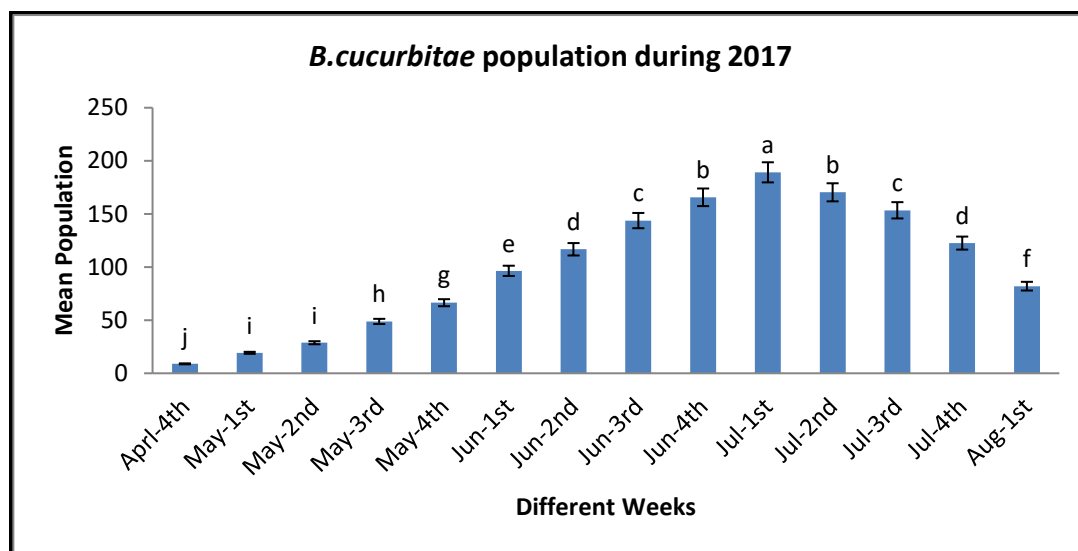


Figure 2. Population of *B.cucurbitae* in bitter gourd recorded on different dates during 2017.

Impact of temperature and humidity on incidence of melon fly: Table 2 showed the relationship of weather factors with activity of melon fly after various intervals application of cue lure in bitter gourd i.e. 15 days, 25 days, 35 days and 45 days had

moderate relationship but also very highly significant positive correlation ($r = 0.4207$; $P < 0.001$) with temperature. While, weak relationship but also highly significant relationship ($r = 0.3309$; $P < 0.001$) was also noted with relative humidity.

Table 2. Correlation of distribution of *B.cucurbitae* (Coq.) with environmental (a biotic) factors.

A biotic factors	r value
Temperature °C	$r = 0.4207$ $P < 0.001$
Relative Humidity (%)	$r = 0.3309$ $P < 0.001$

CONCLUSION

The findings of this study concluded that optimum trapping of melon fruit fly *B. cucurbitae* in bitter gourd was found when cue- lure was changed after 25 days interval whereas, minimum trapping was observed when cue-lure was changed after 45days interval. While, the environmental factors such as temperature and relative humidity (%) had highly significant positive correlation with population activities of flies.

AUTHORS CONTRIBUTION

Muhammad Ibrahim Kubar, wrote the article and provided technical assistance. Bashir Ahmed Kalhoro, conducted experiments, Niaz Hussain Khuhro contributed cue lure & other materials. Muhammad Nawaz Kandhro and Muhammad Saifal Jogi helped in analysis of data,

CONFLICT OF INTEREST.

Authors have no conflict of interest

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